Shovan Shakya

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Education

University of South Florida — **Tampa, Florida**
Bachelors of Science in Electrical Engineering, 3.5 GPA — *May 2025*

Skills

Languages: C, C++, Python, Verilog
Software Development Tools: Zehpyr RTOS, OpenCV, CUDA, Vivado, ROS2, CMake, Git
CAD Tools: Altium, KiCAD, LTSpice, Keysight ADS
Hardware: Oscilloscope, VNA/VSA, Spectrum Analyzer, Logic Analyzer
Certificates: Keysight Technologies RF and Microwave Level 1

Experience

REEKON Tools — **Boston, Massachusetts**
Embedded Software Engineer — *June 2025 - Current*

- Developed new firmware features for the T1R Range Kickstarter launch (above $300K raised), building upon the previous Zephyr RTOS digital tape firmware to integrate LDM and IMU functionality.
- Developed UART driver via bitbanging for Bosch LDM, I2C driver for GPIO Expander and Buck Charger.
- Utilized semaphores to execute tear free LCD update from interrupt.
- Developing Kalman Filter based distance estimation using IMU to auto annotate measurements.
- Developed a camera-based object tracking system using the Goertzel algorithm to isolate LED blink frequencies in video streams, and wrote CUDA kernels to parallelize Goertzel processing across pixels for real-time performance.

Jaycon Systems — **Palm Bay, Florida**
Firmware Engineering Intern — *September 2024 - May 2025*

- Enhanced STM32 firmware in C by reducing SPI EMI for FCC compliance, adding ADC based temperature logging at 2 Hz, and integrating SPI double buffering to prevent DMA buffer overwrites.
- Exposed ADC (thermistor, VIN) and GPIO data to desktop app via C++ and SWIG-generated C# bindings.
- Optimized scheduler parameters, increasing RFID read rates (10 Hz) preserving high performance LED control (45 Hz).
- Architected a Zynq SoC image processing state estimation solution for a NASA SBIR application.
- Developed and tested a connected component labeling (CCL) algorithm in C for ARM Cortex-A9 processor to extract target blob coordinates at a 15 Hz update rate from 720p camera.
- Using High Level Synthesis (C++) to implement FPGA accelerated realtime image processing on Zynq 7000 SoC.
- Developing a CCL algorithm in SystemVerilog and integrated dual-port BRAM to transfer pixel coordinate data from custom IP to the Zynq processor in C++.

Universal Creative — **Orlando, Florida**
Sensor Fusion Intern — *May 2024 - August 2024*

- Developed STM32 firmware in C for IMU interfacing over I2C, communicating with other STM32 modules via CANBUS, and transfer IMU data to a computer using UDP via Ethernet SPI.
- Designed a 4-layer STM32-based PCB (2cm x 2cm) with BNO088 IMU, Ethernet SPI module and TJA1051 CAN IC, supporting up to 30V input voltage.
- Developed a Python tool for synchronized pose and camera data capture on a NVIDIA Jetson based ROS2 platform.
- Created and tested a 1D Bidirectional CNN model for hand gesture recognition using pose data.
- Applied image processing techniques to detect cutouts on display screens to detect bugs and verify manufacture quality.

Monterey Bay Aquarium Research Institute — **Moss Landing, California**
Computer Vision Intern — *June 2023 - August 2023*

- Developed OpenCV based realtime disparity and distance estimation tool for fisheye stereo cameras with accuracy range of 5 meters (+- 300 mm) for desktop and VR.
- Used multiprocessing to perform disparity and tracking in parallel.
- Integrated FathomNet YOLOv5 deep sea organism tracking model with distance estimation.
- Used Sockets to send realtime frames, and distance information between Python OpenCV application and Unity VR.

Projects

Dynamic Bandpass Filter for Voice Modulation using NRF52 — **2025**

- Implemented DPPI/Timer-triggered ADC sampling for mic, reducing CPU involvement.
- Utilized Arm CMSIS DSP library for FFT calculation of ADC data.
- Designing LVGL based TFT screen visulizer for frequency spectrum.